MATH 202A: Introduction to Topology and Analysis

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11/9/26 Lecture 30: Quotients, Products, and Compactness

Recap: When given a sub-basis, we can make a topology using unions of finite intersections of elements of the sub-basis. When given a basis, we can do the same thing with just unions.

Moving onto homeomorphisms, consider the following example.

Example: A Continuous Bijection That Isn't a Homeomorphism

Let $f : [0, 1) \to S^1$ by $f(x) = e^{2\pi i x}$. f is continuous and bijective, but its inverse is not continuous. This is because $[0, 0.5)$ is open, but the arc $[1, -1) \subseteq S^1$ is not.

Quotient Topology

Let $(X, \mathcal{T})$ be a topological space and $\sim$ an equivalence relation on X. Now, consider

$$X/\sim = \{[x] : x \in X\}$$

We know there exists a canonical quotient map $q : X \to X/\sim$ by $x \mapsto [x]$. We define the topology on $X/\sim$ to be given by

$$U \subseteq X/\sim \text{ is open } \iff q^{-1}(U) \text{ is open in } X$$

Example: $\mathbb{R}/\mathbb{Z}$

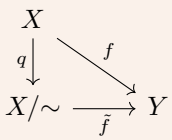
Consider $\mathbb{R}$ with $x \sim y \iff x - y \in \mathbb{Z}$. Then, we see

$$\mathbb{R}/\sim = \{[x] : x \in [0, 1)\}$$

Another useful way to think of it is $[0, 1]$ but we identify 0 and 1. Note that this gives $[0, 1)$ a topology different from the subspace topology, since $[0, 0.5)$ is not open in this topology.

Proposition 30.1

Let X be a topological space with equivalence relation $\sim$. Let $f : X \to Y$ be continuous and constant on all equivalence classes. Then, there exists a continuous map $\tilde{f} : X/\sim \to Y$ such that $f = \tilde{f} \circ q$.



Proof: Let $\tilde{f}([x]) = f(x)$, which is well defined as f is constant on each equivalence class. Now, if U is open in Y, then

$$q^{-1}(\tilde{f}^{-1}(U)) = f^{-1}(U)$$

which is open as f is continuous, so $\tilde{f}^{-1}(U)$ is open in $X/\sim$, since $q^{-1}(V)$ is open iff V is open.

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Example: $\mathbb{R}/\mathbb{Z} \cong S^1$

If $f(x) = e^{2\pi i x}$, it respects $x \sim y \iff x - y \in \mathbb{Z}$. Thus, we have a map $\tilde{f} : \mathbb{R}/\mathbb{Z} \to S^1$ that turns out to be a homeomorphism.

Product Topology

Let $X = \prod_{\alpha \in I} X_\alpha$ where each X_α is a topological space. We want to define a topology on X. To motivate, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is generated by the basis $U \times V$ ($U \subseteq \mathbb{R}$, $V \subseteq \mathbb{R}$ open).

Definition 30.2: Box Topology

The *box topology* on X is generated by the basis

$$\left\{ \prod_\alpha U_\alpha \mid U_\alpha \subseteq X_\alpha \text{ is open} \right\}$$

This is not ideal, however, and we will show why.

Example: The Issue with the Box Topology

Say we have $f_\alpha : Y \to X_\alpha$. We can form $f : Y \to X$ by $f(y) = \{f_\alpha(y)\}_{\alpha \in I}$. One desired property to want is that f_α continuous $\implies$ f continuous. It turns out the box topology doesn't give this.

Let $X = \prod_{n=1}^\infty \mathbb{R}$ with the box topology. Let $f_n(x) = x$, $f_n : \mathbb{R} \to \mathbb{R}$. Now, this gives us a map $f : \mathbb{R} \to X$ by

$$f(x) = (x, x, \ldots)$$

It turns out this is not continuous! Consider

$$U = (-1, 1) \times \left(-\frac{1}{2}, \frac{1}{2}\right) \times \left(-\frac{1}{3}, \frac{1}{3}\right) \times \cdots$$

This is open, but

$$f^{-1}(U) = \{0\}$$

so f is not continuous!

Now, given $X = \prod_\alpha X_\alpha$, we get a map $\pi_\alpha : X \to X_\alpha$ by $\pi_\alpha(x) = x(\alpha) := x_\alpha$.

Definition 30.3: Product Topology

The *product topology* is the topology generated by the sub-basis

$$\bigcup_{\alpha\in I}\{\pi_\alpha^{-1}(U) : U \subseteq X_\alpha \text{ open}\}$$

Note

$$\pi_\alpha^{-1}(U) = U \times \prod_{\alpha'\neq\alpha} X_{\alpha'}$$

This topology can also be described as being generated by the basis

$$\left\{\prod_\alpha U_\alpha : U_\alpha \subseteq X_\alpha \text{ is open and equal to } X_\alpha \text{ for all but finitely many } \alpha\right\}$$

Note that the product and box topologies are the same on finite products. However, in general, the box topology is larger, or "finer".

Definition 30.4: Finer and Coarser Topologies

Let $\mathcal{T}$ and $\mathcal{T}'$ be topologies on the same space X. Suppose $\mathcal{T} \subseteq \mathcal{T}'$. We say $\mathcal{T}'$ is *finer* than $\mathcal{T}$ and $\mathcal{T}$ is *coarser* than $\mathcal{T}'$.

Proposition 30.5

Let $f : A \to \prod_\alpha X_\alpha$ such that $f(a) = (f_\alpha(a))_\alpha$. Then, f is continuous with respect to the product topology iff f_α is for all α.

Proof: If f is continuous, then as $f_\alpha = \pi_\alpha \circ f$, we see f_α is continuous.

Conversely, if f_α is continuous for all α, let

$$U = U_{\alpha_1} \times \cdots \times U_{\alpha_n} \times \prod_{\alpha\neq\alpha_i} X_\alpha$$

be a basis element. Then,

$$f^{-1}(U) = \bigcap_{i=1}^n f_{\alpha_i}^{-1}(U_{\alpha_i})$$

which is open. Thus, as the preimages of basis elements are open, f is continuous.

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Compactness

Definition 30.6: Open Covers

Let $(X, \mathcal{T})$ be a topological space. An *open cover* of X is a collection $\{U_\alpha\}$ of open sets such that $X = \bigcup_\alpha U_\alpha$. A *finite subcover* is a finite subcollection $\{U_{\alpha_i}\}_{i=1}^n$ such that $X = \bigcup_{i=1}^n U_{\alpha_i}$.

Definition 30.7: Compactness

A topological space $(X, \mathcal{T})$ is *compact* if every open cover has a finite subcover.

Note: If $A \subseteq X$, A is compact in the subspace topology iff

$$A \subseteq \bigcup_\alpha U_\alpha \implies A \subseteq \bigcup_{i=1}^n U_{\alpha_i}$$

Proposition 30.8

A closed subset of a compact set is compact.

Proof: Let $A \subseteq X$ be closed. Then, if $\{U_\alpha\}$ is an open cover of A, then $\{U_\alpha\} \cup \{A^c\}$ is an open cover of X. Now, we get a finite subcover

$$X = \bigcup_{i=1}^n V_{\alpha_i} \implies A \subseteq \bigcup_{i=1}^n V_{\alpha_i}$$

where V_{α_i} is drawn from $\{U_\alpha\} \cup \{A^c\}$. Removing A^c from the collection, we still cover A with a finite subcollection of $\{U_\alpha\}$.

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